

Motion control and power coordination of electric propulsion and braking distributed on multiple axles on heavy vehicles

Sachin Janardhanan^{*†}, Leo Laine^{*†}, Mats Jonasson^{*}, Bengt Jacobson^{*}, and Esteban Gelso[†]

Abstract—Instantaneous allocation of power in a distributed electric powertrain reduces losses by selective usage of actuators. Such an allocation also poses a challenge of coordinating power among actuators optimally. This study explores three different approaches of motion control and power coordination of electric machines and service brakes, using an instantaneous power allocation. These approaches based on control allocation framework, are further evaluated and compared for a given use case using simulations. The distinctive strength of the power loss minimisation approach, by using regressed data of power losses from electric machine models instantaneously, is highlighted. Additionally, it is also shown that the power loss minimisation approach resulted in the lowest energy loss in actuator power losses compared to the other approaches.

I. INTRODUCTION

The propulsion system of long haul battery electric heavy vehicles (BEVs), in specific the 35 tonne tractor-semitrailer, requires around 500 kW of continuous electric power to achieve an adequate performance in all operating conditions. However, to decelerate such a vehicle during normal driving, surplus power is required from the service brakes in addition to what the electric machines can provide. For example, the deceleration of 0.25g at a cruising speed of 85 km/h requires more than 2000 kW on a levelled road. Hence, optimal allocation of actuator usage is essential to reduce the power losses for improved efficiency and performance. Additionally, distributed powertrain with drives on multiple axles provide the opportunity to blend the usage of electric machines among the axles along with service brakes [1]–[3]. However, the previous study in [3] using control allocation (CA), is limited to the coordination of power between electric machines on different axles and excludes service brakes.

For an over-actuated system, CA framework distributes total control effort, generalised as forces and moments, among a redundant set of actuators while adhering to constraints. CA frameworks have been previously solved using QP in different applications [4]–[8], here named as CAQP. CAQP problem formulations are in general application-specific and the objective functions for the optimisation along with constraints can be expressed in different ways [5], [8]–[11]. Implementation of CA on electric ground vehicles has shown

promising results [12], [13]. Previous literature on CAQP and comparisons of the different optimisation techniques exist for the flight applications [6], [14]. However, there are limited studies available on comparisons of the different CAQP problem formulations for the BEVs.

The most commonly used CAQP problem formulation in the literature is the mixed optimisation formulation [4]–[7], [9]. The strength of the mixed optimisation formulation is that the vehicle's motion control requests, associated with global forces and moments, are solved in the same cost function that allows balancing between vehicle stability and desired actuator inputs, for example, due to energy efficiency. This is especially preferred if the vehicle is driving in limiting conditions (e.g. low friction) where all the global forces and moments can't be fulfilled. The drawback with the mixed optimisation formulation is that the cost function is a balance between optimising the motion control requests and the desired actuator inputs, where the motion control requests usually becomes the dominant part [9]. Instead of solving the CAQP with motion control requests in cost function, one can also add an equality constraint to represent the global forces and moments that shall be fulfilled. However, this formulation requires that the global forces and moments are part of a feasible solution that the actuators can satisfy within their capabilities. This allows the cost function to be used only for minimising the power losses of the actuators while the equality constraint guarantees the motion, as shown in [10], [8]. Both [10] and [8], are limited to light vehicles and use simulations to perform the evaluations. In [10], the power loss minimisation scheme uses experimental measurements of the electric machines to synthesise power losses. Furthermore, in both the previous studies, the power losses from the service brakes are excluded in the optimisation.

This paper extends the power loss minimisation method studied in [10] and [8], by coordinating the power between different classes of electric machines and the service brakes. Furthermore, we improve the optimisation by using regressed data of power losses from the physical models of the actuators instantaneously, for the current actuator speed state. The performance of the power loss minimisation method is then compared with two different variations of the mixed optimisation formulation for a distributed powertrain based BEV.

II. SYSTEM OVERVIEW AND MODELLING

Three different CA based motion control and power coordination problem formulations are analysed and compared using simulations. For further discussions the term

This work was supported by Energy Agency in Sweden, part of Vinnova/FFI.

^{*}Sachin Janardhanan, Leo Laine, Mats Jonasson, and Bengt Jacobson, are with Chalmers University of Technology, Gothenburg, Sweden {sachin.janardhanan, leo.laine, mats.jonasson, bengt.jacobson}@chalmers.se

[†] Sachin Janardhanan, Leo Laine and Esteban Gelso are with Volvo Group, Gothenburg, Sweden {sachin.janardhanan, leo.laine, esteban.gelso}@volvo.com

'approach' is used to represent the term 'problem formulation' unless explicitly stated. The model of the system used to analyse the different approaches is shown in Fig. 1. The system is modelled and simulated in a python based development environment and the time-integration of the equations of motion are performed using the forward Euler scheme.

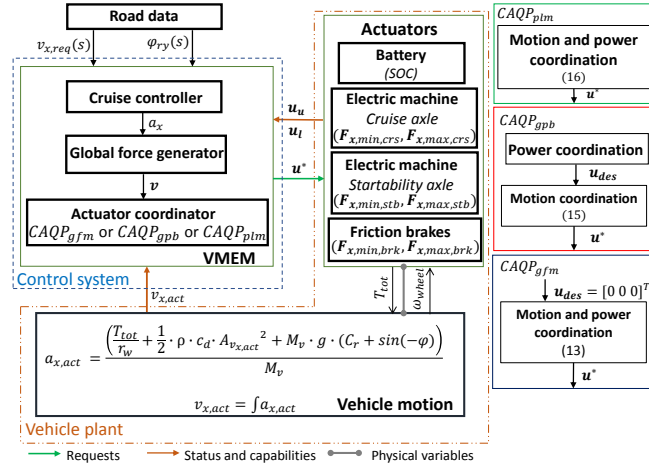


Fig. 1. System model based on control allocation framework used for simulation. The actuator coordinator runs different motion control and power coordination approaches keeping the same interface.

A. Vehicle motion

Vehicle motion subsystem represents the vehicle plant dynamics of the system excluding the actuators. It includes the fundamental relations representing longitudinal vehicle motion modelled as a two-axle tractor-semitrailer with drive on both axles using the relevant parameters as in Table I. The front axle is configured as the cruise axle and rear axle as the startability axle with single gear ratios on each axle along with service brakes. The vehicle is also modelled as a single lumped mass to reduce the influence of high frequency dynamics effects due to load transfer, tyre slip, etc. Such a vehicle model assists in prioritising the comparison of the different approaches without much ambiguity on high frequency vehicle dynamics effects. Furthermore, this simplification also allows the forward Euler scheme to have a higher integration time step. An integration time step of $t_{step} = 1.5s$, is therefore chosen, which permits to capture the steady state vehicle motion dynamics without longer simulation times.

A simple test manoeuvre is used to evaluate the system response and compare the approaches. The velocity profile to be tracked as a function of distance travelled (S) along the straight road stretch with no grade is shown in the Fig. 2. The profile consists of straight line acceleration and braking manoeuvres, that occur during normal driving situations. As seen in the figure, two deceleration events are performed, the first one with $-0.1g$ and the second one with $-0.3g$ approximately.

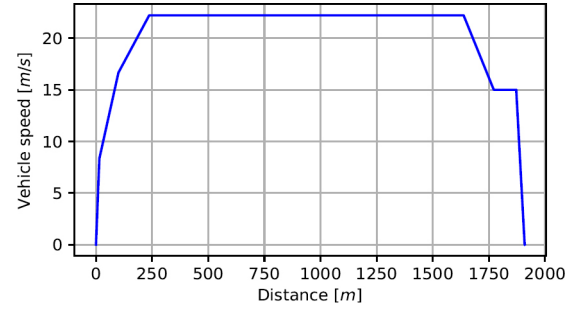


Fig. 2. Test manoeuvre - velocity profile as a function of distance.

TABLE I
VEHICLE AND ENVIRONMENT PARAMETERS

Parameters	value
Maximum vehicle speed [km/h], $v_{x,max}$	90
Vehicle target startability grade [rad], $-\varphi_{start}$	0.08
Vehicle mass [kg], M_v	$36 \cdot 10^3$
Frontal area [m ²], A_v	9
Rolling resistance coefficient [-], C_r	0.005
Drag coefficient [-], C_d	0.59
Wheel radius [m], r	0.506
Air density [kg/m ³], ρ	1.2
Gravity constant [m/s ²], g	9.81
Max. electric propulsion power [kW], P_{max}	600
Battery capacity [kWh], E_{batt}	570
Battery efficiency [-], η_{rt}	0.9
Initial State of charge of the Battery [%], SoC_{init}	100
Road grade [rad], φ_{ry}	0
Cruise axle gear ratio [-], gr_{crs}	12
Cruise axle max. continuous power [kW], $P_{max,crs}$	300
Cruise axle max. rotational speed [rpm], $N_{max,crs}$	15000
Startability axle gear ratio [-], gr_{stb}	23
Startability axle max. continuous power [kW], $P_{max,stb}$	300
Startability axle max. rotation speed [rpm], $N_{max,stb}$	15000
Max. service brake torque [Nm], $T_{lim,brk}$	$200 \cdot 10^3$

B. Vehicle motion control and energy management

The vehicle motion control and energy management (VMEM) subsystem with the signal interfaces as in Fig.1 represents the control architecture. The VMEM consists of cruise controller, global force generator and actuator coordinator functions. The vehicle state estimation functionality is not considered in this study to avoid complexity and the ideal vehicle's motion state variables are used to emulate the states.

1) *Cruise controller*: A simple proportional controller tracks the actual vehicle velocity against the reference velocity request and calculate the necessary acceleration request $a_{x,req}$ according to (1).

$$a_{x,req} = \frac{(v_{x,req} - v_{x,act})}{t_{acc}} \quad (1)$$

To achieve acceptable tracking, the desired time constant is set to $t_{acc} = 1.5s$ for the closed loop controller. The actual acceleration $a_{x,act}$, as in (2), requested by the controller, is then regulated not to exceed the acceleration capability of the vehicle $a_{x,cap}$ calculated from the global force generator.

$$a_x = \text{sgn}(a_{x,req}) \cdot (\min(|a_{x,req}|, a_{x,cap})) \quad (2)$$

2) *Global force generator*: The global force generator instantaneously predicts the maximum possible global force to achieve the required longitudinal vehicle motion. In this paper, only forward driving is considered. The total propulsion force ($F_{x,req}$) needed to accelerate the vehicle and compensate for the rolling, air drag and grade resistances, also called road loads ($F_{x,res}$), is given by (3).

$$F_{x,req} = M_v \cdot a_x + F_{x,res} \quad (3)$$

$$F_{x,res} = \frac{1}{2} \cdot \rho \cdot C_d \cdot A_v \cdot v_x^2 + M_v \cdot g \cdot (C_r + \sin(-\varphi_{ry})) \quad (4)$$

A propulsion force limit ($F_{x,lim}$) is then calculated to limit $F_{x,req}$, so that it does not exceed physical limits considering the coefficient of friction (μ) at the tyre road contact and actuator capabilities. In this evaluation, the coefficient of friction is assumed to be high, $\mu = 1$.

$$F_{x,lim} = \begin{cases} 0.7 \cdot g \cdot M_v, & \text{if } a_{x,req} < 0 \\ \min(M_v \cdot g \cdot (\sin(-\varphi_{start})), \frac{P_{max}}{\max(1, v_{x,act})}) & \text{otherwise} \end{cases} \quad (5)$$

Using $F_{x,lim}$ and $F_{x,res}$ the acceleration capability $a_{x,cap}$ of the vehicle is calculated as to ensure that no acceleration request more than the capability limit is requested.

$$a_{x,cap} = \begin{cases} \frac{(F_{x,lim} - F_{x,res})}{M_v}, & \text{if } |F_{x,lim}| > |F_{x,res}| \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

$$v = \begin{cases} \max(F_{x,req}, \sum_{i=1}^n u_i(i)), & \text{if } F_{x,req} < 0 \\ \min(F_{x,req}, \sum_{i=1}^n u_i(i)), & \text{if } F_{x,req} \geq 0 \end{cases} \quad (7)$$

The global force request (v) requested by the global force generator is then limited as (7). The maximum and minimum force capabilities, $u_u = [F_{x,max,crs}, F_{x,max,stb}, F_{x,max,brk}]^T$ and $u_l = [F_{x,min,crs}, F_{x,min,stb}, F_{x,min,brk}]^T$ of the electric machines and service brakes are used to define the actuator limits in CAQP and act as box constraints for the optimisation. These actuator limits are calculated according to (9). For electric machines, these capabilities are computed using torque limits for a given machine speed, as shown in Fig.3, is a point-to-set mapping of which maximum and minimum is computed.

3) *Actuator coordinator*: The main objective of the actuator coordinator functionality is to map the global force request v to actuator requests $u = [F_{x,crs}, F_{x,stb}, F_{x,brk}]^T$, where $F_{x,crs}$ is the cruise axle force, $F_{x,stb}$ the startability axle force and $F_{x,brk}$ the brake force. The coordination of global force request (v), based on the CA scheme, as actuator requests, is achieved using quadratic optimisation instantaneously. The optimisation is performed using the information on configuration, capabilities and status of the actuators in addition to the request v . The different approaches of mapping the global force requests to the actuator requests

are detailed in III coordination. Using the common actuator coordinator interface, the system is simulated using different approaches at a time.

C. Actuators

The actuator subsystem consists of simple physical and control models of axles and the battery. Each axle model in turn includes a model of an electric machine, transmission and service brake. The characteristics of electric machines are implemented as lookup tables of torque and angular speed, as shown in Fig. 3. These look up tables are used to find the efficiencies, torque limits and power losses. The data for these lookup tables are extracted from electric machine models and can also be obtained at each time step. The efficiency for a single gear ratio based transmission on each axle is simplified as η_{trmn} , and is assumed as a function of the gear ratio.

$$\eta_{trmn,i} = 0.99^{\frac{gr_i}{3}} \quad \text{where } i = crs, stb \quad (8)$$

The service brakes are simplified and lumped into one equiv-

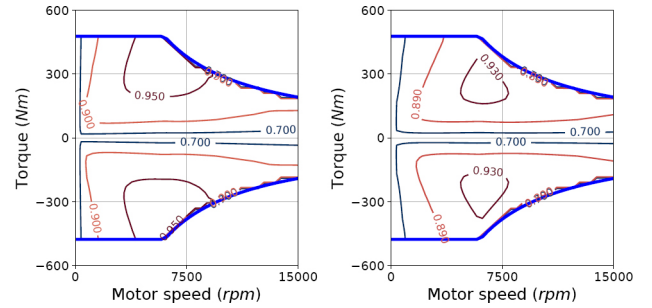


Fig. 3. Efficiency contours (labeled contour lines with efficiencies within the blue lines) and torque limit ($T_{lim,i}$ where $i = crs, stb$), represented by thick blue line, of 300 kW electric machines; (left): Electric machine on cruise axle; (right): Electric machine on startability axle.

alent force for the whole vehicle. This simplification avoids complexities of brake force distribution due to load transfer and tyre slip considerations. Furthermore, only the actuation part of the brakes are considered and losses in the pneumatic system and supply are excluded. These capabilities, as in (9) are then fed to the actuator coordinator functionality layer.

$$F_{x,min,i} = \begin{cases} \min(T_{lim,j}(\frac{v_x \cdot gr_j}{r})) & j = crs, stb \\ -\frac{T_{lim,k}}{r}, & k = brk \end{cases} \quad (9)$$

$$F_{x,max,i} = \begin{cases} \max(T_{lim,j}(\frac{v_x \cdot gr_j}{r})) \\ 0, & i = crs, stb, brk \end{cases}$$

Furthermore, the actuator requests are then converted into actuator forces from each actuator, which are send as requests to the vehicle plant. The controller within the actuators also receives the status of vehicle speed, v_x , for closed loop control. All the three approaches are configured with the same actuator specifications.

Finally, the efficiency at which battery power is consumed $PBatt_{cons}$ and regenerated $PBatt_{gen}$, also including inverter efficiency, are assumed to be the same and simplified to a single value, called battery efficiency η_{rt} as in (10).

$$PBatt_{cons} = \frac{P_{EM}}{\eta_{rt}}, \quad PBatt_{gen} = P_{EM} \cdot \eta_{rt} \quad (10)$$

III. MOTION CONTROL AND POWER COORDINATION APPROACHES

The three different approaches of motion control and power coordination for a BEV with a distributed powertrain is studied using the CAQP scheme. The CA maps the actuator inputs ($\mathbf{u} \in R^m$) to the virtual global force requests ($\mathbf{v} \in R^k$ where $k < m$) using a nonlinear function $g(\cdot)$.

$$\mathbf{v} = g(\mathbf{u}(t)) \quad (11)$$

For instantaneous allocation using CA usually a linear mapping performs satisfactorily and hence (11) can be rewritten with actuator constraints as:

$$\mathbf{v} = B \cdot (\mathbf{u}(t)), \text{ where } B \in R^{k \times m} \quad (12)$$

$$\mathbf{u}_l \leq \mathbf{u}(t) \leq \mathbf{u}_u$$

where B is called the control effectiveness matrix and is set as $B = [1, 1, 1]$ for the approaches, corresponding to the number actuator requests. In the context of distributed drive concept, the global longitudinal force request $\mathbf{v}$ (also referred to as virtual control input in CA) is to be instantaneously coordinated as a cruise axle force ($F_{x,cruise}$), startability axle force ($F_{x,stab}$) and brake force ($F_{x,brk}$) while minimising power consumption. Using this framework of CA, three approaches of the motion control and power coordination are numerically solved using QP. The system modelled, using (1)-(10), is used to evaluate the response for each approach.

A. Global force error minimisation (CAQP_{gfm})

The global force error minimisation, CAQP_{gfm}, approach performs the tasks of motion control and power coordination simultaneously. The CAQP_{gfm} approach is based on the mixed optimisation problem formulation in (13). The term $\|B\mathbf{u} - \mathbf{v}\|$ minimises the allocation or the global force error, needed to achieve the desired vehicle motion. Similarly, the term $\|\mathbf{u}_{des} - \mathbf{u}\|$ minimises the desired control effort error. The desired control effort, $\mathbf{u}_{des}$, is not controlled in this approach directly and is set to $\mathbf{u}_{des} = [0, 0, 0]^T$, which itself acts as a set point. The term γ is the weighting factor to prioritise between the two terms in (13) and in the case of CAQP_{gfm} the global force error minimisation is always prioritised by setting $\gamma > 0$.

$$\mathbf{u}^* = \arg \min (\|W_u(\mathbf{u}_{des} - \mathbf{u})\|_2^2 + \gamma \|W_v(B\mathbf{u} - \mathbf{v})\|_2^2)$$

$$\text{s.t. } \mathbf{u}_l \leq \mathbf{u} \leq \mathbf{u}_u \quad (13)$$

$$\mathbf{u}_{des} = [0, 0, 0]^T$$

The parameters used in the CAQP_{gfm} problem formulation as in (13) are shown in Table II. The actuator weighting matrix (W_u) is used to prioritise the usage of a certain actuator, scalar weighting factor (γ) to prioritise between

TABLE II
PARAMETERS USED IN CAQP_{gfm} AND CAQP_{gpb} APPROACHES

Parameters	CAQP _{gfm}	CAQP _{gpb}
W_u [-]	diag ($10^{-4}, 10^{-1}, 10$)	diag ($10^{-1}, 10^{-1}, 10^{-1}$)
W_v [-]	[1]	[1]
γ [-]	1000	1000

global force term and pure actuator request term and a scalar global force-weighting term W_v to prioritise among global forces to be minimised. In this study the W_u parameter is set so as to prioritise the cruise axle usage always, which generally has the higher efficiency, followed by startability axle and brakes, respectively. This is achieved by setting lower weights for corresponding entry of cruise axle usage in W_u . Setting higher values to an actuator penalises the corresponding actuator. This method of prioritisation is sometimes an advantage in cases with prior knowledge of efficiencies, usage of same type of machines and a lesser number of actuators. Furthermore, to represents the cases of discrete control of using clutch or electrically disconnecting the electric machines, this approach is suitable. In contrast, this approach also puts demand on prioritisation tuning of the actuators, W_u , especially when there are more number actuators and global forces to control. This in turn may lead to sub-optimal solutions.

The mixed optimisation formulation (13) is then solved by reformulating it as a QP problem as in (14).

$$\min_{\mathbf{u}} \quad \frac{1}{2} \mathbf{u}^T H \mathbf{u} + \mathbf{g}^T \mathbf{u} \quad (14)$$

$$\text{s.t. } \mathbf{u}_l \leq \mathbf{u} \leq \mathbf{u}_u$$

$$\text{where } H = 2(W_u^T W_u + \gamma B^T W_v^T W_v B),$$

$$\mathbf{g}^T = -2(\mathbf{u}_d^T W_u^T W_u + \gamma \mathbf{v}^T W_v^T W_v B)$$

The solution is generally seen to be sensitive to W_u , γ and the virtual input values $\mathbf{v}$ itself. This approach is independent of the input from the power coordinator or energy management layer inputs. A possible method to solve this would be to make W_u an adapting parameter by some law through, a power coordinator or upper energy management layers. But the translation of W_u to a adaptive parameter or a variable requires efforts of offline mapping, tuning and validation for the solution stability.

B. Global force error and power loss balancing (CAQP_{gpb})

In conditions when, the electric machines are overheated, fault mode occurs and the regenerative capacity associated with the battery is saturated, the previous approach with a fixed configuration of W_u and $\mathbf{u}_{des} = [0, 0, 0]^T$ may not be efficient. The global force error and power loss balancing approach, CAQP_{gpb}, in such cases would be beneficial by distributing effort among other actuators like brakes or even following predictive actuator usage from the upper software layers so that the performance of electric machines are never limited. Thus, the (CAQP_{gpb}), an extension to the CAQP_{gfm} prepares the desired coordinated actuator

requests for the power coordination separately from the motion control and solved simultaneously. The control layout of the $CAQP_{gpb}$ approach is shown in Fig.1 and the problem formulation in (15).

$$\begin{aligned} \mathbf{u}^* &= \arg \min (\|W_u(\mathbf{u}_{des} - \mathbf{u})\|_2^2 + \gamma \|W_v(B\mathbf{u} - \mathbf{v})\|_2^2) \\ \text{s.t. } \mathbf{u}_l &\leq \mathbf{u} \leq \mathbf{u}_u \end{aligned} \quad (15)$$

The desired control effort is envisioned to be synthesised based on the energy cost of using the actuators from the energy management layers, actuator limits and global force requests. In this study, the desired control effort $\mathbf{u}_{des}$ is calculated so that the sum of all the desired requests is equal to the global force requests. This ensures that the the problem is well-posed and solver is stable. A pseudocode of the $\mathbf{u}_{des}$ calculation for the use-case in this study is formulated as shown in Algorithm 1. In this approach, the parameter W_u

Algorithm 1 Algorithm to compute $\mathbf{u}_{des}$ based on $\mathbf{v}$

```

if  $v \leq 0$  then
  if  $v \leq F_{x,min,crs}$  then
     $\mathbf{u}_{des}(1) = F_{x,min,crs}$ 
     $\mathbf{u}_{des}(2) = \max(F_{x,min,stb}, v - F_{x,min,crs})$ 
    if  $v < (F_{x,min,crs} + F_{x,min,stb})$  then
       $\mathbf{u}_{des}(3) = v - (F_{x,min,crs} + F_{x,min,stb})$ 
    else
       $\mathbf{u}_{des}(3) = 0$ 
    end if
  else if  $v > F_{x,min,crs}$  then
     $\mathbf{u}_{des}(1) = v$ ;  $\mathbf{u}_{des}(2) = 0$ ;  $\mathbf{u}_{des}(3) = 0$ 
  end if
else if  $v > 0$  then
  if  $v \leq F_{x,max,crs}$  then
     $\mathbf{u}_{des}(1) = v \cdot 0.7$ ;  $\mathbf{u}_{des}(2) = v \cdot 0.3$ ;  $\mathbf{u}_{des}(3) = 0$ 
  else if  $v > F_{x,max,crs}$  then
     $\mathbf{u}_{des}(1) = F_{x,max,crs}$ ;  $\mathbf{u}_{des}(3) = 0$ 
     $\mathbf{u}_{des}(2) = \min(F_{x,max,stb}, v - F_{x,max,crs})$ 
  end if
end if

```

as in Table II, is set equally because the actuator requests are coordinated before the optimisation algorithm and therefore specific actuators need not be prioritised.

C. Power loss minimisation ($CAQP_{plm}$)

Another approach to the optimal motion control and power coordination is by obtaining an analytical solution to a constrained quadratic problem, as in [8] and [10]. Both the power and motion control coordination is performed simultaneously. For a lower order CA problem, as in the use case of this study, the quadratic formulation with a linear equality constraint ensures that the problem is convex and there exists a global minimum. The objective in this approach is to minimise instantaneous power losses from electric machines and service brakes while the motion control is assured with the help of equality constraints $B \cdot \mathbf{u} = \mathbf{v}$.

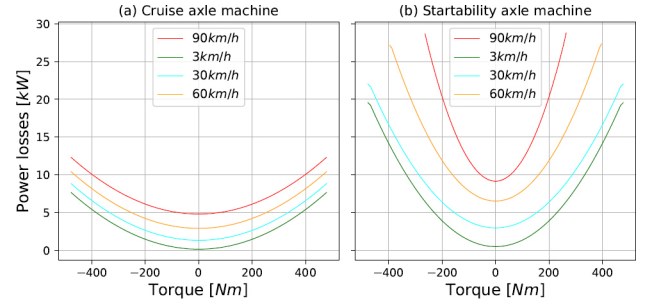


Fig. 4. Power losses of electric machines as a function of torque for different vehicle speeds. The offset of power losses at zero torque is referred to as idle losses which is higher at higher speeds.

The optimisation problem therefore can be expressed as:

$$\begin{aligned} \min_{\mathbf{u}} \quad & (Ploss_{crs} + Ploss_{stb} + Ploss_{brk}) \\ \text{s.t. } \quad & B\mathbf{u}^T = \mathbf{v} \\ & \mathbf{u}_l \leq \mathbf{u} \leq \mathbf{u}_u \end{aligned} \quad (16)$$

The heat losses from the electric machines and service brakes are the only form of losses considered. Power losses in other forms like tyre slip, rolling resistance, etc., are excluded in the scope of this study with the aim to have a uniform comparison with the other approaches.

1) *Electric machine losses:* Analytically, the heat loss from the electric machines in its simplest form can be expressed as a quadratic function of torque:

$$Ploss_{heat} = \frac{R}{(K_{torq})^2} \cdot T_{em}^2 \quad (17)$$

where R is the total inner resistance and K_{torq} is the motor constant of the electric machine. However, in this study, the power loss characteristics are extracted from a lookup table as a function of torque and speed. These lookup table data are based on the models of electric machines and evaluated using copper, iron and windage losses. The total power losses obtained for each machine type, from the look up tables, as a function of their operating torque range for different machine speeds are seen in Fig. 4. These loss models, also include a loss at zero torque, referred to as idle losses. Upon close observation, it is seen that the power loss curves of the electric machines can also be approximated using a quadratic regression function, analogous to (17). The polynomial expression for a regressed quadratic approximation of power losses can then be expressed as:

$$Ploss_i \approx a_i \cdot T_i^2 + b_i \cdot T_i + c_i, \quad (18)$$

where, a_i, b_i and c_i - curve fitting coefficients,
 $i = crs, stb$

To get the fitting curve for the quadratic function, the entire operating torque range is used for both the electric machines. This approximation also well suits the minimisation of the power losses using the quadratic optimisation.

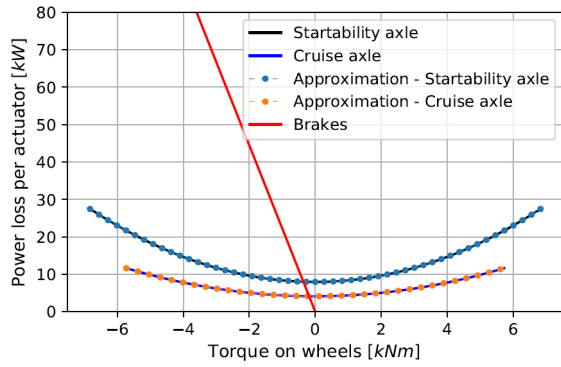


Fig. 5. Power losses of electric machines with their quadratic approximations and service brakes, as a function of wheel torque for a vehicle speed of 80km/h. The curve fit parameters for the regressed quadratic approximations of: a) cruise axle electric machine - $a_{crs} = 0.033$, $b_{crs} = 0.0022$, $c_{crs} = 4130.44$ giving an R^2 fit of 0.99 b) Startability axle electric machine - $a_{stb} = 0.21$, $b_{stb} = -0.031$, $c_{stb} = 7779.22$ giving an R^2 fit of 0.99.

2) *Service brake losses*: The service brake usage is associated with 100% loss, unlike the electric machines, which are also used for recuperation of the energy. Therefore, losses from the service brakes can then be approximated by the simple linear relation of wheel power as:

$$P_{loss_{brk}} = -T_{brk} \cdot \omega_{whl} \quad (19)$$

where ω_{whl} is the wheel speed.

However, to include the contribution of service brakes into the QP formulation and to be expressed using a quadratic polynomial, as in (18), a numerically small coefficient $a_{brk} = 1e^{-5}$ is used as the coefficient of the quadratic term. This small value, instead of zero, is chosen to make the hessian matrix for the QP positive definite and thereby solve the system.

Additionally, to include the difference in the type of actuator (electric machines and brakes) and their operating range, the torque at the wheel is used as a common reference scale to compare the power loss contribution of each actuator. Fig. 5, shows the power losses and quadratic approximation of individual actuators as a function of wheel torque for a given vehicle speed. The optimisation problem in (16), using (18) and (19), can now be reformulated into a standard quadratic optimisation form as:

$$\min_{\mathbf{u}} \quad \frac{1}{2} \mathbf{u}^T H \mathbf{u} + g^T \mathbf{u} \quad (20)$$

$$\text{s.t. } B \cdot \mathbf{u} = \mathbf{v}$$

$$\mathbf{u}_l \leq \mathbf{u} \leq \mathbf{u}_u$$

$$\text{where } H = 2 \cdot \begin{bmatrix} \frac{r^2}{gr_{crs}^2} \cdot a_{crs} & 0 & 0 \\ 0 & \frac{r^2}{gr_{stb}^2} \cdot a_{stb} & 0 \\ 0 & 0 & a_{brk} \end{bmatrix}$$

$$g^T = \left[\frac{r}{gr_{crs}} \cdot b_{crs}, \frac{r}{gr_{stb}} \cdot b_{stb}, -\omega_{whl} \cdot r \right]$$

In a standard QP formulation, the constant term, here in this study the term c_i representing idle losses of the actuators at

zero torque, is neglected as they do not add any information to the optimisation.

IV. RESULTS AND DISCUSSIONS

Energy consumption metrics were used to compare the motion control and power coordination approaches. The total energy consumed for the test manoeuvre is calculated as a function of the rate of change of the battery state of charge (SoC) over the distance travelled.

$$E_{cons, SoC} = E_{batt} \cdot 3.6 \cdot 10^6 \cdot \left(1 - \frac{SoC_{end}}{100}\right), \quad (21)$$

$$\text{where, } SoC_{end} = 100 - \left(\int_0^S d(SoC)\right)$$

The total energy consumed, $E_{cons, SoC}$, from the battery for the three approaches is shown in Table III. $E_{cons, SoC}$ include the energy lost to overcome the motion resistance, power losses in the actuators and conversion losses due to efficiencies of the battery and gears. As seen, the $CAQP_{gpb}$ approach has the highest energy consumption and the $CAQP_{plm}$ approach the lowest. The differences in

TABLE III
ENERGY CONSUMPTION AND LOSSES FOR ACTUATOR COORDINATION APPROACHES

Motion control and power coordination approach	Change of SoC from start to end in the test manoeuvre, ($E_{cons, SoC}$) [MJ]	Energy loss in all the actuators (Electric machines and Brakes), ($E_{cons, PL}$) [MJ]	Energy loss in all the actuators, excluding the idle losses (c_i) term for the electric machines, ($E_{cons, PL-QA}$) [MJ]
$CAQP_{gfm}$	88.89	6.87	5.87
$CAQP_{gpb}$	89.56	6.84	5.82
$CAQP_{plm}$	89.51	6.79	5.77

$E_{cons, SoC}$ for the different approaches are further inspected by observing how the actuators are coordinated during the test manoeuvre. Fig. 6, shows the force usage of the actuators for the different approaches during the test manoeuvre. The distinct differences between the approaches are evident for the segments of test manoeuvre with the lower acceleration, i.e., for the distance from 500m to 1600m. In the $CAQP_{gfm}$ approach, the heuristic method of prioritising the usage of the most efficient machine, i.e. the cruise axle electric machine, is clearly seen. The cruise axle machine usage is maximised throughout the manoeuvre and supplemented by startability axle machines whenever needed. The $CAQP_{plm}$ approach in contrast utilises both the cruise axle and startability axle machines, with a significant force proportion coordinated to the startability axle. $CAQP_{gpb}$ approach also utilises the cruise and startability axles, with actuator forces lying in between the levels of the other two approaches. Thus, observing the results of $E_{cons, SoC}$ the $CAQP_{gfm}$ qualifies as the most optimal approach.

However, comparing $E_{cons, PL}$ the trend is opposite. As seen in Table III, $E_{cons, PL}$ for the $CAQP_{plm}$ approach is the optimal one with minimal energy consumption because

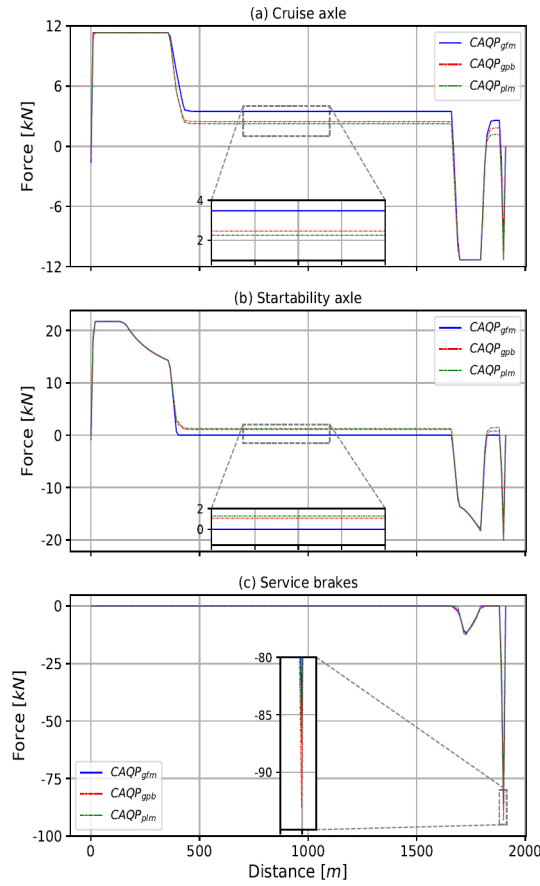


Fig. 6. Actuator coordination for different approaches. Each subplot shows the usage of respective actuators for different approaches. For example the subplot (a) shows the usage of cruise axle for the three different approaches and similarly subplot (b) and (c) for startability axle and brakes respectively. The x-axis in the last subplot is common for the other subplots.

of lower power losses, which is expected. Fig. 7 also shows the energy loss contribution by each actuator, because of the power losses for the different approaches. The sum of energy losses due to the contribution from each actuator for the use case is seen in Fig. 7. This corresponds to the $E_{cons,PL}$ results in the Table III for each approach. Therefore, from these results it is clearly seen that the $CAQP_{gfm}$ is not optimum in the case of $E_{cons,PL}$. Furthermore, it is also seen that the $CAQP_{gpb}$ approach through selective actuator coordination of 70-30% distribution between the cruise and start axle, respectively, also reduces power losses.

To analyse the differences in energy consumption trends between $E_{cons,SoC}$ and $E_{cons,PL}$, an analysis of the vehicle performance measures for the different approaches is performed. Table IV, presents the root mean square (rms) longitudinal vehicle acceleration and time taken to complete the test manoeuvre for the approaches. As noticeable, the vehicle motion is not the same among all approaches. The $CAQP_{gpb}$ approach has the highest value in both the metrics of rms acceleration and time taken, compared with the other two approaches. This also corresponds to the highest $E_{cons,SoC}$ value for the $CAQP_{gpb}$ approach in Table

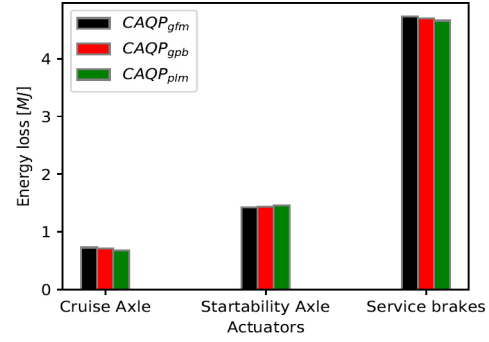


Fig. 7. Energy loss in the actuator due to power losses from each actuator for the different approaches for the given use case.

III. Similarly, the difference in rms acceleration for the $CAQP_{plm}$ and $CAQP_{gfm}$ approaches explains the trend of $E_{cons,SoC}$ values. Additionally, to investigate the differences

TABLE IV
VEHICLE MOTION PERFORMANCE

Approach	Root mean square acceleration [m/s^2]	Time taken to complete the test manoeuvre in simulation [s]
$CAQP_{gfm}$	0.464	145.5
$CAQP_{gpb}$	0.4760	147
$CAQP_{plm}$	0.4711	145.5

in vehicle motion measures, further analysis is performed by observing the profiles of the acceleration difference between the approaches, as shown in Fig. 8. The major differences among the approaches, are seen during the deceleration phase of the test manoeuvre. It is seen that the $CAQP_{gfm}$ approach has the highest value of difference in acceleration between each of the other two approaches. This justifies the lower rms acceleration results for the $CAQP_{gfm}$ approach in the Table IV, which also implies that the other two approaches have higher longitudinal acceleration. Similarly, the phase difference of acceleration difference plots explains the additional time taken to complete the manoeuvre using the $CAQP_{gpb}$ approach. These differences in acceleration profiles is attributed to the formulation of different approaches and thereby the solution obtained. For e.g., to achieve the motion control in the $CAQP_{plm}$ approach, the control requests to the actuators are achieved using the equality constraint, $B \cdot u = v$ and therefore tries to strictly follow the motion request. Alternatively, in the $CAQP_{gfm}$ approach, the motion control requests are achieved by minimising the error of the term $\|Bu - v\|$. This leads to extremely small differences in the force allocation among the actuators and similarly differences in the vehicle acceleration. Lastly, the motion in $CAQP_{gpb}$ approach, tries to follow the $CAQP_{gfm}$ approach but deviates due to the additional u_{des} inputs and therefore cannot exactly follow the minimisation of the motion term $\|Bu - v\|$ error. Since, the motion between the approaches is not the same, it is difficult to conclude

on the most optimal approach. This analysis indicates that the results would be different if a longer or different route is chosen for the study. Therefore, it can be concluded that an combined analysis of both the energy consumption and motion performance is needed to evaluate the performance of different approaches.

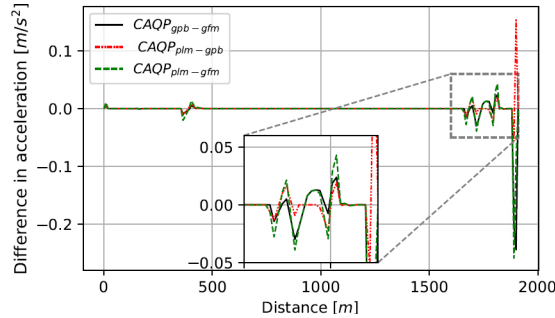


Fig. 8. Actual acceleration difference along the distance travelled between the different approaches. Black solid line: $CAQP_{gpb} - CAQP_{gfm}$; red dashdotted line: $CAQP_{plm} - CAQP_{gpb}$; green dashed line: $CAQP_{plm} - CAQP_{gfm}$.

An additional measure of energy losses in the actuators, ($E_{cons, PL-QA}$), if the actuator plant models used the quadratic approximation without the c_i term as in (18), is shown in the Table III. The results of $E_{cons, PL-QA}$ also confirm that the $CAQP_{plm}$ approach provides the optimal solution of actuator coordination with minimal power losses. Furthermore, the difference between $E_{cons, PL-QA}$ and $E_{cons, PL}$ represents the idle losses, c_i term, of the electric machines. This measure is highlighted here to show the further potential of energy minimisation possible, if the idle losses of the electric machines are included in the optimisation. Nonetheless, this constant term representing idle losses is not included in the standard quadratic optimisation. This becomes important when there is an opportunity to use a clutch to disconnect the electric machine from usage, as shown in [10]. Excluding the idle losses (c_i) term in the quadratic optimisation, as in the current $CAQP_{plm}$ approach, will result in higher energy losses since the optimisation method always considers two machines to coordinate power. In other words the standard quadratic optimisation cannot distinguish the use of a clutch and can end up with non optimal solutions. Therefore, $CAQP_{plm}$ approach, which uses quadratic optimisation in its current form is limited for the cases when one of the electric machines is not used. In such cases, the $CAQP_{gfm}$ and $CAQP_{gpb}$ approaches have the advantage by actively prioritising the electric machines to be used, indirectly representing the use of a clutch. However, the problem of tuning and prioritising actuators in the $CAQP_{gfm}$ and $CAQP_{gpb}$ needs careful treatment and may not be optimal in terms of power losses in the actuator.

V. CONCLUSIONS

In this study three approaches of the motion control and power coordination based on control allocation framework,

namely global force minimisation ($CAQP_{gfm}$), global force and power balance minimisation ($CAQP_{gpb}$) and power loss minimisation ($CAQP_{plm}$), are analysed and compared for a specific use case using simulations. Metrics of total energy consumption, energy consumption because of power losses in the actuators and vehicle performance measures are used to compare the approaches. The $CAQP_{plm}$ approach results in minimal energy consumption because of the lower power losses in actuators. Additionally, the simulation results also indicate that the motion between the approaches is not the same. These differences are because of the inaccuracies of achieving motion requests due to the different problem formulations. Further, the limitations of the $CAQP_{plm}$ approach, in particular the ability of the standard QP formulation, of not handling the idle losses of electric machines in the optimisation is highlighted. This limitation is the scope for future work using mixed integer programming.

REFERENCES

- [1] C. Satzger, R. de Castro, and T. Bunte, "A model predictive control allocation approach to hybrid braking of electric vehicles," in *2014 IEEE Intelligent Vehicles Symposium Proceedings*, 2014, pp. 286–292.
- [2] T. Bächle, K. Graichen, M. Buchholz, and K. Dietmayer, "Model predictive control allocation in electric vehicle drive trains," *IFAC-PapersOnLine*, vol. 48, no. 15, pp. 335–340, 2015, 4th IFAC Workshop on Engine and Powertrain Control, Simulation and Modeling E-COSM 2015. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S2405896315019242>
- [3] S. Janardhanan, L. Laine, M. Jonasson, B. Jacobson, and M. Alaküla, "Concept design of electric cruise and startability axles for long haul heavy vehicles to maximise driving range," in *2021 IEEE Vehicle Power and Propulsion Conference (VPPC)*, 2021, pp. 1–8.
- [4] O. Härkegård, "Dynamic control allocation using constrained quadratic programming," in *Proceedings of the 2002 AIAA Guidance, Navigation, and Control Conference and Exhibit*, 2002. [Online]. Available: <http://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-55978>
- [5] J. Plumlee, D. Bevely, and A. Hodel, "Control of a ground vehicle using quadratic programming based control allocation techniques," in *Proceedings of the 2004 American Control Conference*, vol. 5, 2004, pp. 4704–4709 vol.5.
- [6] J. Petersen and M. Bodson, "Constrained quadratic programming techniques for control allocation," *IEEE Transactions on Control Systems Technology*, vol. 14, no. 1, pp. 91–98, 2006.
- [7] A. Sinigaglia, K. Tagesson, P. Falcone, and B. Jacobson, "Coordination of motion actuators in heavy vehicles using model predictive control allocation," in *2016 IEEE Intelligent Vehicles Symposium (IV)*, 2016, pp. 590–596.
- [8] A. Pennycott, L. D. Novellis, P. Gruber, A. Sornioti, and T. Goggia, "Enhancing the energy efficiency of fully electric vehicles via the minimization of motor power losses," in *2013 IEEE International Conference on Systems, Man, and Cybernetics*, 2013, pp. 4167–4172.
- [9] L. Laine, "Reconfigurable motion control systems for over-actuated road vehicles," Ph.D. dissertation, 06 2007.
- [10] J. Torinsson, M. Jonasson, D. Yang, and B. Jacobson, "Energy reduction by power loss minimisation through wheel torque allocation in electric vehicles: a simulation-based approach," *Vehicle System Dynamics*, vol. 0, no. 0, pp. 1–24, 2020. [Online]. Available: <https://doi.org/10.1080/00423114.2020.1858121>
- [11] C. Schreppel and J. Brembeck, "A qp solver implementation for embedded systems applied to control allocation," *Computation*, vol. 8, no. 4, 2020. [Online]. Available: <https://www.mdpi.com/2079-3197/8/4/88>
- [12] Y. Chen and J. Wang, "Energy-efficient control allocation for over-actuated systems with electric vehicle applications," in *Dynamic Systems and Control Conference*, vol. 44175, 2010, pp. 37–44.
- [13] R. J. P. de Castro, "Motion control and energy management of electric vehicles," Ph.D. dissertation, Universidade do Porto (Portugal), 2013.
- [14] M. Bodson, "Evaluation of optimization methods for control allocation," *Journal of Guidance, Control, and Dynamics*, vol. 25, no. 4, pp. 703–711, 2002.